Multi-Vector Vision-Language Models for Complex Visual Document Search

Dr. E. Thorne, Dr. A. Vance — Deep Multimodal Retrieval Lab

Abstract — Conventional OCR pipelines discard multi-column layout geometry, font hierarchies, and tabular alignments. In this paper, we demonstrate that visual late-interaction MaxSim embeddings preserve diagrammatic context while delivering sub-millisecond retrieval across dense industrial PDF corpora.

1. Introduction
Visual document retrieval replaces brittle textual extraction with direct patch embeddings. Documents such as circuit schematics, pinout charts, and audited financial statements communicate vital relationships through 2D spatial arrangement rather than linear token streams.

2. Empirical Benchmarks
We evaluated ColPali and ColQwen on engineering datasheets. As detailed in Table 1, visual late interaction achieved 0.892 NDCG@5, outperforming text-only BM25 (0.512) and single-vector CLIP (0.641).

Table 1: Retrieval Performance on Engineering Datasheets Benchmark

Model Representation	NDCG@5	MRR@10	Recall@5	Index Size (MB)
BM25 (Tesseract OCR)	0.512	0.480	0.620	12.4
Dense CLIP (ViT-L/14)	0.641	0.615	0.710	48.2
ColPali (PaliGemma-3B)	0.865	0.832	0.915	184.0
ColQwen2.5 (3B Dynamic)	0.892	0.868	0.942	192.5
ColQwen2.5 INT8 Quantized	0.884	0.860	0.938	96.2
ColQwen2.5 1-Bit Binary	0.861	0.835	0.910	12.0